

Scleral necrosis after radiation therapy for uveal melanomas: report of 23 cases

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Abstract

Aims To describe the evolution and management over time of scleral necrosis as a complication after radiation therapy for uveal melanomas.

Design Retrospective nonrandomized case series.

Methods Twenty-three patients (12 male, 11 female; median age 56.4 years; range, 30–73 years) treated with a single plaque therapy (cobalt-60 (Co^{60}) or iodine-125 (I^{125}) (1.5% of treated patients) or proton beam radiotherapy (0.45% of treated patients) for choroidal or ciliary body melanoma, presenting scleral necrosis as a post-radiation complication, were studied. Tumor characteristics, intraocular pressure, location and size of scleral necrosis were recorded during visits after radiation therapy.

Results The median maximal tumour base and thickness at diagnosis were 14.6 ± 4.4 mm and 7.2 ± 2.3 mm respectively.

Median radiation dose to tumor base and tumor apex was 223 Gy and 88.3 Gy respectively. The mean time of scleral necrosis onset after radiation therapy was 70.4 months (range 11–257 months); the mean scleral necrosis diameter at its onset was 4.4 mm (range 1–10 mm). Out of 23 affected eyes, 17 presented a stability of scleral necrosis; in three cases there was partial regression of scleral necrosis; in two cases there was a progressive extension of scleral necrosis until a pre-perforation stage. In two cases, a secondary enucleation was necessary due to a severe neovascular glaucoma.

Conclusions Scleral necrosis is a rare complication of radiotherapy for uveal melanomas. Relapse with extra-scleral extension of the primary tumor has to be ruled out in order to avoid unnecessary enucleations. In most patients, the scleral necrosis remains stable and does not need any further treatment.

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Introduction

Plaque radiotherapy and proton beam radiotherapy are the standard treatment for ciliary body and choroidal melanoma. At the Institut Curie, plaque radiotherapy with cobalt-60 isotope was used till the end of the 1980s. Iodine-125 plaques were then used to treat all uveal melanomas between the end of 1989 and 1991. Since 1991, iodine-125 plaques have been employed for small anterior tumors, and proton beam was used for all other tumors. The radiotherapy side effects may include cataract, radiation retinopathy, neovascular glaucoma, vitreous haemorrhage and scleral necrosis. Scleral necrosis is an uncommon complication of radiation therapy for uveal melanoma [1,

2]. Scleral thinning as well as increased transparency occurs, but staphylomas are rare. Scleral necrosis is characterized by the almost total lack of any symptoms, and sometimes it is accompanied by overlying deposits of pigment. As the primary tumor is brought under control, the necrotic areas are absorbed or sequestered, leaving an ectasia with underlying uvea exposed or covered with a thin film of conjunctiva or episclera. All cases of scleral necrosis in patients with choroidal melanoma that have been described in the literature were caused by plaque radiotherapy; none has been described after proton beam therapy [3–6].

Even though this complication has been already described, its rarity makes it often under diagnosed, the main and most difficult differential diagnosis being a relapse of the initial tumour with extrascleral extension.

In this report we describe the scleral necrosis after radiotherapy for choroidal and ciliary body melanoma its evolution and management over time.

Materials and methods

Iodine plaque therapy for choroidal melanoma is considered as a standard treatment in our institution, approved by the institutional review board. All the patients treated by proton beam irradiation from 1991 to 1993 were treated according to a prospective therapeutic protocol and signed an informed consent. From 1993, the protons were considered as a standard treatment in our institution, approved by the institutional review board, and these patients do not need to sign a separate consent. Follow-up examinations of all patients treated conservatively (including intraocular press-ion) are generally made at 6-month intervals up to 2 years and every 12 months thereafter.

All patients treated at the Institut Curie are informed that the data from their medical charts may be used anonymously, and sign an authorization at their first visit. The database is declared according to the French legislation to the Commission Nationale Informatique et Libertés (CNIL), the administration office that authorizes and regulates the use of electronic databases.

The database of our institution was reviewed in order to identify all cases of scleral necrosis after radiation therapy for uveal melanoma. Eyes presenting extrascleral extension at diagnosis were excluded from this study. Among the 3,631 patients in the database, 23 were identified as having scleral necrosis (0.63%).

Diagnosis of scleral necrosis was made on the basis of the clinical appearance of a pigmented scleral area overlying the tumour base and after a relapse of the tumour had been excluded. A colour photograph of the lesion was realised at diagnosis and during follow-up.

A retrospective review of the clinical records was carried out. Patient data included at initial examination age, gender (male, female), involved eye, medical problems (none, diabetes mellitus, hypertension), associated ocular pathologies.

Tumor data were obtained for each patient: these were (1) anatomic location of the uveal melanoma (iris, ciliary body and choroids), (2) meridian location of tumor epicentre (superior, superotemporal, temporal, inferotemporal, inferior, inferonasal, nasal, superonasal), (3) posterior tumor margin (posterior to equator, between equator, anterior to equator), (4) maximum basal tumor diameter and thickness (based on ultrasonography in millimetres) at diagnosis and at time of scleral necrosis appearance, (5) shape (dome, mushroom), and (6) theoretical volume (calculated with the formula $(4/3 \pi d^2 h)/2$, in which d is the tumor diameter divided by 2 and h is tumor thickness) at diagnosis and at time of scleral necrosis appearance [2, 7].

The plaque radiotherapy data were the radioactive isotope employed (iodine-125, cobalt-60), the plaque size, the radiation dose (Gy) to the tumor apex and tumor base. The plaque size was selected to be 2 mm larger than the base of the entire intraocular tumor on all margins.

The proton beam radiotherapy data were the radiation dose to the tumor apex and tumor base: cobalt-60 Gray equivalent divided into four fractions of 15 Gy on 4 days.

With regard to the scleral necrosis we noted the following details: (1) date of onset, (2) size and meridian location of scleral lesion epicentre (superior, superotemporal, temporal, inferotemporal, inferior, inferonasal, nasal, superonasal), (3) instrumental (magnetic resonance imaging or ultrasound biomicroscopy) or histopathologic (biopsy) reports when present, and (4) evolution during follow-up.

A descriptive analysis of these data has been realised.

Results

The median age at diagnosis of the 23 patients (12 male, 11 female; nine right eyes, 14 left eyes) examined was 56.4 years (range, 30–73 years). Six patients had systemic hypertension, one patient had diabetes mellitus, one had rheumatoid polyarthritis and one developed a myelodysplasia syndrome. The remaining patients did not have any systemic disorder.

Initial tumour characteristics are described in Table 1, the majority of patients presented a ciliary body melanoma (Table 1).

The primary tumor posterior margin was posterior to equator in two cases, at the equator in four cases and anterior to equator in 17 cases; all tumours had at least part of the lesion located anterior to the equator.

The median intraocular pressure at scleral necrosis onset was 19.6 mmHg (range 8–40 mmHg); topical hypotensive therapy was prescribed in seven patients. In only three of them, the intraocular pressure after treatment was <18 mmHg.

Table 1 Tumor characteristics in the 23 subjects studied

	Number		Mean	Median	Range
	n=23	(%)			
At diagnosis					
Tumor location					
Choroidal	7	30.4			
Ciliochoroidal	9	39.2			
Ciliary body	3	13.0			
Iridociliary	4	17.4			
Tumor meridian					
Superior	2	8.7			
Superotemporal	6	26.2			
Temporal	3	13.0			
Inferotemporal	3	13.0			
Inferior	4	17.4			
Inferonasal	0	0.0			
Nasal	1	4.3			
Superonasal	4	17.4			
Tumor shape					
Dome	15	65.2			
Mushroom	8	34.8			
Tumor diameter (mm)			14.6	14	8–20.9
Tumor thickness (mm)			7.2	6.6	3–11
Tumor volume mm ³)			3946.4	3344.4	401.8–9055.1
At scleral necrosis onset					
Tumor thickness (mm)			3.9	4.0	1–9.2
Tumor volume (mm ³)			2719.8	2215.2	209.3–7857
Δ* tumor thickness (mm)			3.3	2.2	0–10
Δ* tumor volume (%)			39.8	37.2	0–86.8

(*) variation in tumor dimension between diagnosis and scleral necrosis onset

The radioisotope used for plaque radiotherapy was iodine-125 in nine cases out of 585 treated patients during the same period (1.5% of treated patients) and cobalt-60 in three cases. The plaque diameter was 12 mm in two cases, 14 mm in two cases, 15 mm in three cases, 16 mm in four cases and 18 mm in two cases. None of these patients needed a muscle detachment during surgery.

Proton beam radiotherapy was used in ten cases out of 2,239 treated patients (0.45%).

In one case, subsequent iodine-125 plaque therapy was necessary as secondary treatment because of no response to proton beam.

Radiation characteristics are described in Table 2.

Table 2 Radiation characteristics in 23 subjects treated with radiation therapy for uveal melanoma

	Number		Mean	Median	Range
	n=23	%			
Radiation source					
Iodine-125	9	39.2			
Cobalt-80	3	13.0			
Proton beam	10	43.5			
Proton beam/Iodine-125	1	4.3			
Radiation dose (Gy) at tumor apex			88.3	74	56–305
Radiation dose (Gy) at tumor base			223.0	114	60–592
Follow-up after radiotherapy (month)			115.6	103	23–257

Table 3 Scleral necrosis characteristics in 23 subjects treated with radiation therapy for uveal melanoma

	Number		Mean	Median	Range
	n=23	%			
SN size (mm)					
Cobalt-60			3.3	3	1–6
Iodine-125			5.1	4	2.5–10
Proton beam			4.3	4.5	1.5–8
Total			4.4	4	1–10
SN meridian					
Superior	1	4.3			
Superotemporal	5	21.8			
Temporal	4	17.4			
Inferotemporal	3	13.0			
Inferior	3	13.0			
Inferonasal	1	4.3			
Nasal	1	4.3			
Superonasal	5	21.8			
SN evolution					
Stability	17	74			
Progression*	2	8.7			
Perforation†	1	4.3			
Regression	3	13.0			
Enucleation‡	2	8.7			
Time free of SN (month) by radiation source:					
Cobalt-60			177	178	97–257
Iodine-125			49	41	23–111
Proton beam			62.4	61.5	11–107
Total			70.4	60	11–257

SN: scleral necrosis

* Until pre-perforation

† Post-traumatic evolution

‡ Secondary enucleation for severe neovascular glaucoma

The characteristics of scleral necrosis are described in Table 3. All scleral necrosis (100%) appeared in the meridian coinciding with and adjacent to the tumor center.

In nine cases, clinical appearance and ultrasound excluded a relapse with extrascleral extension. For 14 patients, further instrumental examination was needed for differential diagnosis of primary tumor recurrence with extrascleral extension. Ultrasound biomicroscopy was used in five cases, and magnetic resonance imaging in six cases.

In three cases, a biopsy of the suspect pigmented lesion was performed. The biopsy showed tumor cells in none of the patients, but only macrophagic deposits.

During follow-up after the first description of the scleral necrosis, none of these patients needed any treatment. Three out of 23 patients showed partial regression of the scleral necrosis, 17 patients (73.9%) showed no change or very faint and slow progression (Fig. 1). In two out of 23 cases, the scleral necrosis increased to a pre-perforation status. In one case, a post-traumatic perforation at the site of scleral necrosis was documented. The patient healed spontaneously after topical antibiotics.

In two cases a secondary enucleation was required for severe neovascular glaucoma [8]. The histopathologic reports of the two enucleated eyes confirmed the presence of scleral necrosis.

Discussion

Radiation therapy is the standard conservative treatment of uveal melanoma; it may lead to several early and late complications, including scleral necrosis [1].

Even though already described after plaque radiotherapy, diagnosis of scleral necrosis and its management is not familiar to the majority of ophthalmologists. And it is often diagnosed as extraocular extension.

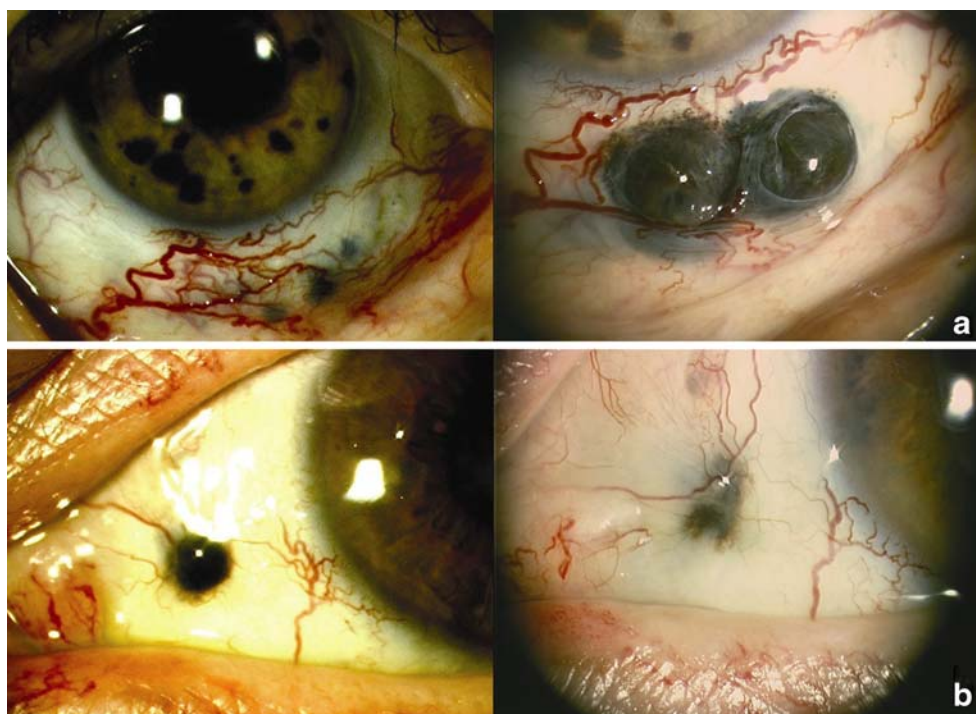
In 1999, Gündüz et al. described scleral necrosis in 11% of 136 patients treated with plaque radiotherapy (iodine-125) for uveal melanomas with predominantly ciliary body involvement, with a mean time to the development of scleral necrosis of 27 months (median 32 months, range 4–120 months). Intraocular pressure (>15 mmHg) and tumor thickness (>7 mm) were the most significant variables predictive of scleral necrosis [3].

In 2002, Shields et al. [4] reported scleral necrosis in 7% at 5 years and in 9% at 10 years in 354 consecutive patients treated with plaque radiotherapy (iodine 125) for large posterior uveal melanomas (≥ 8 mm thickness).

In contrast to these studies, Conway et al. described no case of scleral necrosis at 24 months in 21 patients treated with proton beam radiotherapy for extra-large uveal melanoma [6].

In our study we report 23 cases of scleral necrosis; nine cases (1.5% of treated patients) related to iodine-125 plaque radiotherapy, three related to cobalt-60 plaque radiotherapy

Fig. 1 **a** Initial scleral necrosis (*left*) 6 years after proton beam therapy for uveal melanoma in a 70-year-old patient; progression to a pre-perforation status (*right*) 2 years after the onset of scleral necrosis. **b** Initial scleral necrosis (*left*) 2 years after proton beam therapy for uveal melanoma in a 37-year-old patient; partial regression (*right*) 11 years after the onset of scleral necrosis



and ten cases (0.45%) after proton beam therapy. In one case, a iodine-125 plaque therapy was applied as a second treatment because of no response to proton beam. To our knowledge, scleral necrosis after proton beam therapy has never been described previously. Unfortunately, the low number of reported eyes didn't allow any statistical evaluation on its appearance with respect to radiation source utilised for melanoma therapy.

All patients described in our study presented with a clinically visible scleral necrosis area. The tumor location of our patients was predominantly at the equator or anterior to the equator. Scleral necrosis located posteriorly may be underdiagnosed, as it cannot be easily documented by external examination and ultrasonography. It is possible that a certain number of patients treated for posterior uveal melanoma with proton beam may have a small area of scleral necrosis that can not be detected.

Furthermore, scleral necrosis is a late complication; after plaque radiotherapy with cobalt-60, it appeared at a mean interval of 177 months.

The mean time to the development of scleral necrosis for patients treated with plaque radiotherapy and proton beam is 49 months (range 23–111) and 62.4 months (11–107 months) respectively. In our experience, scleral necrosis developed later than was described by Gündüz et al. [3]. Moreover, after proton beam therapy it seems to appear later than after plaque radiotherapy.

Scleral necrosis may be difficult to differentiate clinically from a tumour recurrence with extraocular extension [9]. Gündüz et al. diagnosed scleral necrosis only when visible by ophthalmologic examination [3]. Ophthalmologic examination or ultrasonography were used by Shields et al. [4].

In this series, the diagnosis was clinical for the majority of patients, supported by the fundus examination and B scan showing the absence of tumor evolution. We performed a biopsy in only three eyes with a very suspicious lesion where a relapse was not ruled out. A melanin deposit associated to a discrete number of melanophages were described. No tumoral cell was reported. The histologic report of the two enucleated eyes for severe neovascular glaucoma revealed the presence in the scleral necrosis area of a discrete number of melanophages charged with melanic pigment. These cells were found within the choroid and sclera, coinciding with the tumor centre. No tumoral cell was found.

Biopsy of scleral lesion is not recommended when there is a clinical and paraclinical evidence of scleral necrosis. In the case of a highly suspicious pigmented scleral lesion after radiotherapy for uveal melanoma, a close follow-up of clinical evaluation and repeated B scan or MRI may be realised to rule out a tumor relapse.

Several hypotheses on physiopathology of scleral necrosis have been reported. Gündüz et al. evoke the radio-induced disruption of normal blood supply, the necrotic tumor cells effect by lysozymatic enzymes release, and the highly frequent scleral involvement of large tumors [3].

Other hypotheses have been proposed by Corrêa [2]: as the local ischemic effect on the sclera induced by surgical procedures or tumor necrosis, or as an occult systemic autoimmune vasculitic disease that acts on a weakened scleral area. Based on the histological report of biopsies and enucleated eyes of our series of patients, scleral necrosis does not seem directly caused by radiation, especially after proton beam treatment where the dose received by the sclera is homogeneous and equivalent to the dose received at the apex of the tumor (60 Gy). The presence of a relatively large necrotic tumor mass induced by radiotherapy may initiate a non-specific inflammatory response; consequently, the chemical mediators and cytotoxic products may cause direct cellular damage and incite scleritis and necrosis near the tumor location.

The evolution is favourable overall. The best management of patients with scleral necrosis is observation; none of the patients described needed any kind of treatment. The patient who presented a perforation of the globe secondary to a violent trauma healed spontaneously after topical antibiotherapy. Slit-lamp examination with a description of the area of necrosis, and photographic recording, allow a correct follow-up in time.

Scleral necrosis is a rare but not eye-threatening complication of radiotherapy of uveal melanomas; the major difficulty is to clearly differentiate it from a recurrence with extrascleral extension of the previously treated uveal melanoma, in order to avoid unnecessary enucleations.

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